Assignment No 7

Aim: Implement SVM Algorithm for Regression.

Theory:

Support Vector Machine (SVM) is a supervised machine learning algorithm that can be used for both **classification and regression**. When SVM is used for predicting continuous numerical values, it is called **Support Vector Regression (SVR)**.

SVR attempts to find a function that predicts the target values while keeping the prediction errors within a specified tolerance level. Instead of trying to fit every data point exactly, SVR tries to find a regression function that provides a good balance between **accuracy and model complexity**.

Support Vector Regression (SVR)

In SVR, the objective is to find a regression line or function such that most of the training data points lie within a predefined **ε (epsilon) margin** around it.

The basic regression function can be represented as:

$$f(x) = w^T x + b$$

Where:

- **f(x)** = predicted output
- **w** = weight vector
- **x** = input features
- **b** = bias/intercept

The **epsilon (ε) tube** defines the acceptable error around the predicted function. Errors within this tube are generally not penalized.

Working of SVR

The main steps involved in SVR are:

1. Collect the dataset.
2. Identify the independent variables and continuous target variable.
3. Preprocess the data.
4. Split the data into training and testing sets.
5. Scale the features because SVM is sensitive to feature magnitude.

6. Select an appropriate SVR kernel.
7. Train the SVR model using the training data.
8. Predict the target values for the test dataset.
9. Evaluate the predictions using regression metrics.

Important Parameters of SVR

1. C (Regularization Parameter)
C controls the trade-off between model complexity and training error.

- Higher C → model tries harder to fit training data.
- Lower C → allows more errors and produces a more flexible model.

2. Epsilon (ε)
Epsilon defines the width of the tolerance tube.

A larger epsilon allows more errors without penalty, while a smaller epsilon makes the model try to fit the data more closely.

3. Kernel
The kernel determines how the input data is transformed or represented.

Common kernels include:

- **Linear Kernel**
- **Polynomial Kernel**
- **RBF (Radial Basis Function) Kernel**
- **Sigmoid Kernel**

The **RBF kernel** is commonly used when the relationship between variables is non-linear.

Kernel Function

For the RBF kernel:

$$K(x, x') = \exp(-\gamma \mid\mid x - x' \mid\mid^2)$$

Here, **γ (gamma)** controls how far the influence of a training data point extends.

Difference Between Linear Regression and SVR

Linear Regression	SVR
Finds the best-fit regression line	Finds a regression function using an epsilon-insensitive margin
Mainly suitable for linear relationships	Can handle non-linear relationships using kernels
Less computationally complex	Can be computationally expensive for large datasets
Usually less sensitive to parameter selection	Requires parameters such as C, epsilon and gamma
Easy to interpret	More difficult to interpret

Evaluation Metrics

SVR can be evaluated using:

- **Mean Absolute Error (MAE)**
- **Mean Squared Error (MSE)**
- **Root Mean Squared Error (RMSE)**
- **R^2 Score**

A lower MAE, MSE, and RMSE generally indicate better predictions, while an **R^2 score closer to 1** indicates better model performance.

Advantages

- Can model both linear and non-linear relationships.
- Effective in high-dimensional datasets.
- Uses kernel functions to handle complex relationships.
- Provides good generalization when parameters are properly selected.
- Less affected by small errors within the epsilon margin.

Limitations

- Feature scaling is generally required.
- Choosing suitable values of C, epsilon, and gamma can be difficult.
- Training can become slow for very large datasets.
- The model is less interpretable than simple Linear Regression.

- Performance depends significantly on the choice of kernel and parameters.

Applications

SVR can be used for:

- House price prediction.
- Stock and financial forecasting.
- Sales prediction.
- Temperature prediction.
- Demand forecasting.
- Energy consumption prediction.
- Predicting continuous values in scientific and engineering applications.

Conclusion

Support Vector Regression (SVR) is an SVM-based supervised learning technique used for predicting continuous numerical values. It uses an **epsilon-insensitive tube** and support vectors to construct an effective regression function. By using suitable kernels such as **RBF**, SVR can model complex and non-linear relationships between variables.

Code:

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.svm import SVR

from sklearn.preprocessing import StandardScaler

X = np.array([[1], [2], [3], [4], [5], [6], [7], [8], [9], [10]])

y = np.array([1.2, 1.9, 3.2, 3.8, 5.1, 5.8, 7.2, 7.9, 9.1, 10.2])

sc_X = StandardScaler()

sc_y = StandardScaler()

X_scaled = sc_X.fit_transform(X)

y_scaled = sc_y.fit_transform(y.reshape(-1, 1))

svr_model = SVR(kernel='rbf', C=10, epsilon=0.2, gamma=0.1)
```

```
svr_model.fit(X_scaled, y_scaled.ravel())

X_test = np.array([[6.5]])

X_test_scaled = sc_X.transform(X_test)

y_pred_scaled = svr_model.predict(X_test_scaled)

y_pred = sc_y.inverse_transform(y_pred_scaled.reshape(-1, 1))

print("Predicted value for X = 6.5:", y_pred[0][0])

plt.scatter(X, y, color='red', label='Original Data')

plt.plot(X, sc_y.inverse_transform(svr_model.predict(X_scaled).reshape(-1, 1)),
        color='blue', label='SVR Prediction')

plt.title('Support Vector Regression (SVR)')

plt.xlabel('X')

plt.ylabel('y')

plt.legend()

plt.show()
```

Output:

